

ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-ASYNC

ORINATION: Prof. Kidd, Instructor, UMBC

PURPOSE OF CHANGE: Cyclic executive task manager

DOCUMENT NUMBER: ECN1.1

DATE OF PUBLICATION: 9/29/2026

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TEAM LEAD: Nyi Thu Kha, UMBC

EXECUTIVE SUMMARY:

This ECN documents the change from the Project 1 (PROJECT-ASYNC) design to a round-robin cyclic executive task manager using a function pointer array. The customer requirements, high-level technical requirements, and hardware configuration are unchanged. The software architecture has been refactored. Here, the main loop dispatches three non-blocking tasks in round-robin order via a function pointer table. Testing requirements have been extended to verify the new task manager behavior. The formal design document has been revised accordingly.

HIGH-LEVEL DESCRIPTION:

Modify PROJECT-ASYNC to use a round-robin cyclic executive task manager based on a function pointer array, without changing the customer or high-level technical requirements.

REASON FOR CHANGE:

Results in a more reliable design. Also reduces labor and design costs for implementing future extensions.

Changed Documents:

9/29/2026 (CMPE 311-ASYNC-LED-NYI-92626 → CMPE311-PROJECT-CE-SDD1.1)
System Design Document for PROJECT-ASYNC:

- **Changed Document Name:** “Async LED Blink Controller” to “Cyclic Executive Task Manager – System Design Document”.

- **Changed Document Reference:** “CMPE 311-ASYNC-LED-NYI-92626” to “CMPE 311-PROJECT-CE-SDD.1.1”.
- **Changed Date of Publication:** 9/27/2026 to 9/29/2026
- **HIGH-LEVEL DESCRIPTION:**
 - **Added:** “using a round-robin cyclic executive task manager” at end.
- **DESCRIPTION:**
 - **Added:** “organized as a round-robin cyclic executive with a function pointer table” after “bare-metal, non-blocking software architecture”.
- **RESULTS SUMMARY:**
 - **Added:** “including those for the cyclic executive” before “were satisfied”.
- **REFERENCE:**
 - **Added:** “[1] Nyi Thu Kha, Async LED Blink Controller — System Design Document, CMPE 311-ASYNC-LED-NYI-92626” as first reference.
- **DEFINITIONS:**
 - **Added:** Cyclic Executive, Round-Robin, Function Pointer Table, Task, Dispatcher
- **ACRONYMS AND ABBREVIATIONS:**
 - **Added:** CE, ECN, ECO, ECR.
- **EDUCATIONAL CONTRAINS:**
 - **Added:** EC.6, EC.6.1, EC.6.1.1, EC.6.2
- **ADDITIONAL DESIGN CONSIDERATIONS:**
 - **Replaced:** items 1-9 was replaced with a revised list. Item 1 now references [1] instead of restating the hardware description. Items 2-4 are new and describe the cyclic executive, tasks, and dispatcher. Item 6 now describes shared state at file scope. Remaining items retained from Project 1.
- **SYSTEM DESIGN:**
 - **Added:** New section. Contains Overview, Software Architecture, Cyclic Executive, Function Pointer Table, Task Desing, Timing Analysis, and Changes from PROJECT-ASYNC. This section did not exist in [1].
- **TESTING AND VALIDATION REQUIREMENTS:**
 - **Added:** T.18, T.19, T.20, T.21, T.22.
 - **Changed:** Section now references [1] for T.0 – T.17 rather than restating them.
- **TESTING RESULTS:**
 - **Changed:** Table 4 now summarizes T.0 – T.17 as a range referencing [1] and lists T.18 – T.22 individually.
- **Changed:** GitHub repository link updated from Project 1 to Project 2.

Added Documents:

- ECN1_1 – This Engineering Change Note, dated 9/29/2026.
- async_led_ce.ino – Cyclic executive source code, dated 9/29/2026.

Removed Documents:

None.

Replaced Documents:

None.

Any Additional Design Considerations:

1. The cyclic executive uses round-robin scheduling with no priorities and no preemption.
2. Each task is non-blocking and returns promptly so that no task starves another.
3. Shared states (interval1, interval2, state1, state2, serialState, inputBuffer, etc.). These are accessible to all tasks at file scope.
4. No locking or critical sections are required. Because each task runs to completion before the next is dispatched, and no task can be interrupted by another task.
5. The function pointer table is declared at file scope and has a compile-time-known size (NUM_TASKS computed with sizeof).
6. No third-party multitasking libraries or RTOS are used so EC.5 remains satisfied.
7. All Project 1 1 customer requirements, high-level technical requirements, educational constraints, hardware, and user-visible behavior are unchanged.